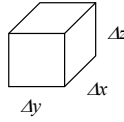


9. Energy Equation

In heat diffusion equations,

$$\dot{E}_{st} = \dot{E}_{in} - \dot{E}_{out} + \dot{E}_g \text{ applied to the CV}$$



$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q}$$

In convection (general case),

- The energy balance $\dot{E}_{st} = \dot{E}_{in} - \dot{E}_{out} + \dot{E}_g$ does not include the work term ($\sim \text{force} \times \text{displacement}$).
- $\dot{E}_{in} - \dot{E}_{out}$ is not only by conduction (diffusion) but also by advection (bulk flow)
- Velocity gradient in viscous flow results in heat dissipation within the CV, i.e., $\dot{q} \neq 0$.

$\Rightarrow$ The conduction equation needs to be generalized, leading to a general energy equation:

$$\begin{aligned} \dot{E}_{st} \rightarrow \rho c_p \frac{\partial T}{\partial t} &= \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) && \leftarrow \dot{E}_{in} - \dot{E}_{out} \text{ by diffusion} \\ &- \rho c_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) && \leftarrow \dot{E}_{in} - \dot{E}_{out} \text{ by advection and pressure work} \\ &+ \mu \Phi && \leftarrow \dot{E}_g \text{ by viscous dissipation} \\ &+ \dot{q} && \leftarrow \dot{E}_g \text{ by other mechanisms} \end{aligned}$$

$$\boxed{\rho c_p \left(\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T \right) = \nabla \cdot (k \nabla T) + \mu \Phi + \dot{q}} \quad (*)$$

<Note>

- ① If $u=v=w=0$ (no flow), energy equation = conduction equation
- ② Viscous dissipation: important for highly viscous or high-speed flow, negligible otherwise

$$\Phi = 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right] + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right)^2$$

$$\Phi = \tau_{ij} \frac{\partial v_i}{\partial x_j} \text{ and } \tau_{ij} = \mu \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right)$$

③ Equation (*) tells you whether a velocity/temperature field can be a solution or not
 $u=10, v=w=0, T=y^2+z-t$ cannot be a solution

④ Eq. (*): linear in T

⑤ The velocity field needs to be known before computing T or T is coupled with the velocity field (T need to be solved simultaneously with $\mathbf{v}$)

A. Mass conservation $\dot{m}_{st} = \dot{m}_{in} - \dot{m}_{out}$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

If steady, $\nabla \cdot (\rho \mathbf{v}) = 0$

If steady and incompressible $\nabla \cdot \mathbf{v} = 0$

B. Momentum conservation $\dot{\mathbf{p}}_{st} = \dot{\mathbf{p}}_{in} - \dot{\mathbf{p}}_{out} + \mathbf{F}$ ($\mathbf{p} = m\mathbf{v}$)

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla p + \rho \mathbf{g} + \mu \nabla^2 \mathbf{v} \text{ for incompressible Newtonian fluid}$$

Newtonian fluid: stress = linear function of strain rate, e.g., $\tau_{xy} = \frac{\partial u}{\partial y}$

C. The mass and momentum conservation equations can be solved without knowing T if incompressible $\rho \neq f(T)$. If $\rho = f(T)$ as in natural convection all three equations are coupled and need to be solved simultaneously.

⑥ Four parameters are involved in determining T (ρ, c_p, μ, k)

$\rho \sim$ Inertia, mass/momentum, energy storage

$c_p \sim$ Thermal inertia, thermal energy storage

$\mu \sim$ Viscous diffusion

$k \sim$ Thermal diffusion

Note the relation between $\nu \equiv \mu/\rho$ and $\alpha \equiv k/\rho c_p$

[Alternative form of energy equation]

$$\rho c_p \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + \mu \Phi + \dot{q},$$

where material (substantial) derivative $\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z} = \frac{\partial}{\partial t} + (\mathbf{v} \cdot \nabla)$

Temperature of a specific particle in Lagrangian description

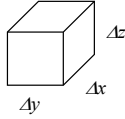
$$T[\mathbf{r}(t), t] = T[x(t), y(t), z(t), t]$$

$$\frac{DT}{Dt} = \frac{\partial T}{\partial t} + \frac{\partial T}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial T}{\partial y} \frac{\partial y}{\partial t} + \frac{\partial T}{\partial z} \frac{\partial z}{\partial t} = u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} = \frac{\partial T}{\partial t} + (\mathbf{v} \cdot \nabla)T$$

Change of T = Local change + Convective change

[Derivation of energy equation]

- Rigorous derivation beyond the course scope
- However, the process is similar to that for conduction eq., continuity eq., etc.
- For example,



$$\dot{E}_{st} = \rho \Delta x \Delta y \Delta z \left(\tilde{u} + \frac{1}{2} v^2 \right) = \rho \Delta x \Delta y \Delta z \left(c_v T + \frac{1}{2} v^2 \right)$$

$$\text{Work by gravity} = \rho \Delta x \Delta y \Delta z \mathbf{g} \cdot \mathbf{v}$$

$$\dot{E}_{in} \text{ by advection at } x = \rho u \Delta y \Delta z \left(\tilde{u} + \frac{1}{2} v^2 \right)_x$$

Rate of work done on the x surface ($\Delta y \Delta z$ surface)

$$\mathbf{F}_x = (p + \tau_{xx})\mathbf{i} + \tau_{xy}\mathbf{j} + \tau_{xz}\mathbf{k}$$

$$\mathbf{v} \cdot \mathbf{F}_x \Delta y \Delta z = [(p + \tau_{xx})u + \tau_{xy}v + \tau_{xz}w] \Delta y \Delta z$$

....

- For details, see Supplementary Material 9SA
- For more details, see Supplementary Material 9SB